Year 12 Maths - Polynomials

Similar questions - Factor theorem and factorising

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question you must show detailed reasoning.

The cubic polynomial $f(x)$ is defined by $f(x) = 3x^3 - 14x^2 + 13x + 6$.

(a) Show that $(x-2)$ is a factor of $f(x)$. **[1]**

(b) Express $f(x)$ in fully factorised form. **[3]**

(c) Hence solve $3 \times 8^y - 14 \times 4^y + 13 \times 2^y + 6 = 0$, giving any exact solutions in terms of logarithms. **[4]**

2 The polynomial $g(x) = 2x^3 + ax^2 - 7x - 6$ has a factor $(x+2)$.

(a) Find the value of a. **[2]**

(b) Hence solve $g(x) = 0$, giving any irrational roots in exact form. **[4]**

(c) Find the set of values of x for which $g(x) > 0$. **[3]**

Question 1

(a) $f(2) = 24 - 56 + 26 + 6 = 0$, so $(x-2)$ is a factor

(b) $f(x) = (x-2)(3x^2 - 8x - 3) = (x-2)(x-3)(3x+1)$

(c) Let $x = 2^y$: $(2^y - 2)(2^y - 3)(3 \times 2^y + 1) = 0$. $2^y = 2 \Rightarrow y = 1$; $2^y = 3 \Rightarrow y = \log_2 3$; $2^y = -\frac{1}{3}$ has no solution. So $y = 1$ or $y = \log_2 3$

Question 2

(a) $g(-2) = -16 + 4a + 14 - 6 = 0 \Rightarrow 4a = 8 \Rightarrow a = 2$

(b) $g(x) = (x+2)(2x^2 - 2x - 3)$. $2x^2 - 2x - 3 = 0 \Rightarrow x = \dfrac{2 \pm \sqrt{4+24}}{4} = \dfrac{1 \pm \sqrt{7}}{2}$. So $x = -2,\ \dfrac{1 \pm \sqrt{7}}{2}$

(c) The cubic has a positive leading coefficient, and its roots in order are $-2 < \dfrac{1-\sqrt{7}}{2} < \dfrac{1+\sqrt{7}}{2}$. So $g(x) > 0$ for $-2 < x < \dfrac{1-\sqrt{7}}{2}$ or $x > \dfrac{1+\sqrt{7}}{2}$